

NDM-CLIP: Normalized Distribution Matching and Masked-Text Prediction for Text-Based Person Search

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Abstract—Text-based person search requires precise alignment between visual appearances and natural language descriptions. While models like CLIP provide strong global representations, they struggle with fine-grained local details and textual noise. To address this, we propose NDM-CLIP, a unified coarse-to-fine framework. First, NDM-CLIP employs a Normalized Image-Text Contrastive loss to stabilize the global feature space. For fine-grained alignment, it utilizes a Gradient-Attention mechanism to dynamically identify and mask noisy tokens. Crucially, we identify that standard forward Similarity Distribution Matching (SDM) is suboptimal under explicit token masking. We therefore introduce a reversed SDM loss that firmly anchors optimization to true positive labels, effectively insulating the training process from mask-induced noise. Combined with Masked-Text Prediction, NDM-CLIP explicitly strengthens local visual-semantic associations. Extensive experiments demonstrate our method’s superiority, particularly under constrained computational resources. Notably, NDM-CLIP achieves 69.76% Rank@1 on the full CUHK-PEDES dataset, significantly outperforming equivalent baselines. Furthermore, it exhibits exceptional efficiency in data-scarce, attaining 51.81% Rank@1 trained on just 10% of CUHK-PEDES and 58.81% Rank@1 on the challenging 3000Vn-V2E benchmark.

I. INTRODUCTION

Text-based person search (TBPS) aims to retrieve pedestrian images based on textual queries, a task that inherently demands simultaneous reasoning about global appearances and fine-grained local attributes. Vision-language models such as CLIP [1] have considerably advanced the field through contrastive learning. However, directly applying CLIP to TBPS uncovers two main weaknesses. First, the standard Image-Text Contrastive (ITC) loss primarily optimizes global representations, compressing the embedding manifold and struggling to capture local identity-specific details. Second, standard CLIP provides no native mechanism to filter noisy tokens or explicitly model token-level visual-semantic associations, resulting in degraded performance when captions are imperfect or training data is scarce.

Recent studies have attempted to tackle these problems, albeit often in isolation. For instance, TBPS-CLIP [2] lessens the distributional imbalance of the standard ITC loss by introducing a Normalized ITC (N-ITC) objective. While effective at stabilizing the global embedding space, this approach acts mainly as a

coarse alignment and remains largely insensitive to fine-grained local features. On the other hand, GA-DMS [3] tackles fine-grained alignment by proposing a Gradient-Attention Similarity Score (GASS) that guides token masking, jointly optimizing a Similarity Distribution Matching (SDM) loss and Masked-Text Prediction (MTP). However, existing SDM objectives traditionally adopt the forward Kullback-Leibler divergence to match the predicted similarity distribution against the target labels. Our empirical analysis reveals that this forward direction is suboptimal under explicit token masking. Because explicit masking artificially introduces noise into the sequence, the forward divergence scatters optimization gradients across numerous false negatives, hindering effective alignment.

Motivated by these observations, we propose **NDM-CLIP**, a unified coarse-to-fine framework that seamlessly integrates global stability with a robust local alignment strategy. In our design, N-ITC is first employed to secure a balanced coarse global manifold. To extract fine-grained details without being overwhelmed by textual noise, we utilize the GASS mechanism as a token-level filter to mask uninformative words. Crucially, to overcome the limitations of standard distribution matching, we introduce a *reversed* SDM formulation. By changing the divergence direction to compute the distance from the target to the prediction, our reversed SDM firmly anchors the optimization to true positive pairs. This ignores the mask-induced noise in negative pairs, empirically yielding more robust alignment. Paired with an MTP head, NDM-CLIP sharply captures local visual-semantic correspondences.

The main contributions of this paper are summarized as follows:

- We present NDM-CLIP, a unified coarse-to-fine framework that integrates coarse global alignment (N-ITC) with a dynamic local token-filtering strategy (GASS) for robust TBPS.
- We empirically demonstrate that the standard forward SDM becomes suboptimal under explicit noise masking. To address this, we propose a reversed SDM formulation that directly anchors gradients to true positive labels, effectively improving alignment robustness.
- Extensive experiments show that NDM-CLIP delivers su-

perior data efficiency, significantly surpassing baselines in 10% of CUHK-PEDES and 3000Vn-V2E dataset.

II. RELATED WORK

A. Advances in Text-Based Person Search

The TBPS landscape has transitioned from unimodal feature extraction to joint vision-language learning. Early methods typically employed CNN-RNN pipelines [4], [5], [6] and often incorporated hand-crafted spatial partitions [7], [8] or external NLP tools [4] to establish fine-grained correspondences. The release of CLIP [1] triggered a paradigm shift, encouraging approaches based on cross-modal transformers for implicit reasoning [9], pseudo-text tokens for pre-training [10], and disordered representation learning [11]. While these CLIP-driven techniques show impressive global alignment, handling identity-level fine-grained distinctions and the inevitable noise in textual descriptions remains a challenge [12], [13].

The rapid progress in TBPS has been powered by breakthroughs in Vision-Language Pre-training (VLP). Foundation models like CLIP build strong joint embedding spaces from massive web-sourced image-text pairs. To improve data efficiency and feature quality, later works have integrated self-supervised objectives [14], token-level similarity measures [15], and multi-view supervision [16]. In parallel, adapting large VLP models to downstream tasks has led to innovations in prompt tuning [17] and multilingual formulations such as mSigLIP [18], successfully applied in low-resource TBPS scenarios [19]. Despite these advances, standard VLP pipelines inherently lack explicit mechanisms to suppress local textual noise during fine-tuning, a gap that calls for specialised optimization strategies.

B. Cross-Modal Optimization and Noise Suppression

Current research increasingly focuses on designing better cross-modal objectives and noise mitigation techniques. The conventional Image-Text Contrastive (ITC) loss, although widely used, is highly susceptible to distributional bias. It treats all negative samples equally and enforces a strict one-to-one matching, which compresses the embedding manifold and restricts representational capacity. To address this, Cao *et al.* [2] introduced the Normalized ITC (N-ITC) loss. N-ITC explicitly stabilizes the global embedding space by leveraging soft labels to balance the target distribution. However, because N-ITC relies heavily on holistic sentence-level aggregation, its primary weakness is that it acts only as a coarse alignment. It remains largely insensitive to fine-grained local features and is particularly vulnerable to uninformative or noisy words inherently present in textual descriptions.

Because global objectives aggregate all words into a single embedding vector, they lack the granularity to isolate and ignore faulty tokens. Consequently, the contrastive mechanism forcefully pushes the entire text embedding away from the correct image representation due to a minor localized error. For example, if a description incorrectly replaces a “blue jacket”

with “red” or includes a hallucinated accessory like a “backpack”, the global loss aggressively shifts the text embedding toward the wrong conceptual cluster in the latent space. This single error can overpower other correctly matched attributes, inevitably leading to retrieval failure. Therefore, integrating local matching mechanisms becomes crucial to explicitly evaluate token-level relevance, ensuring that correctly aligned fine-grained details are emphasized while the disruptive impact of noisy tokens is minimized.

To tackle the challenges of fine-grained alignment and local textual noise, token-level noise suppression mechanisms have received considerable attention. Zheng *et al.* [3] recently proposed the Gradient-Attention Similarity Score (GASS) to dynamically evaluate token importance, enabling selective token masking. They optimized their model by combining Masked-Text Prediction (MTP) and Similarity Distribution Matching (SDM). The MTP objective is fundamentally advantageous because it forces the cross-modal decoder to explicitly reconstruct masked informative tokens using surrounding visual cues. This predictive mechanism explicitly strengthens the local visual-semantic associations that coarse global losses miss.

Meanwhile, the SDM loss is utilized to align the predicted image-text similarity distributions with the ground-truth labels. Nevertheless, existing distribution matching approaches almost exclusively adopt the forward Kullback-Leibler (KL) divergence ($D_{KL}(\mathbf{P}||\mathbf{Q})$). We identify a critical weakness in this forward formulation: it is mathematically suboptimal under explicit token masking. When noisy tokens are explicitly replaced with artificial `[mask]` tokens, unnatural noise is introduced into the sequence. The forward KL divergence forces the optimization gradients to scatter across the entire distribution, including numerous artificially noisy false-negative pairs. This misdirected focus destabilizes the training process and hinders precise alignment. Building on these insights, our work unifies the coarse stability of N-ITC with a mathematically refined, *reversed* SDM formulation ($D_{KL}(\mathbf{Q}||\mathbf{P})$). This reversal exclusively anchors the optimization gradients to the true positive labels, effectively insulating the model from mask-induced noise and achieving both greater robustness and finer alignment.

III. PROPOSED METHOD

We propose NDM-CLIP, a synergistic framework designed to overcome the inherent limitations of standard contrastive learning in TBPS by effectively integrating complementary optimization objectives. The logical flow of our approach (illustrated in Fig. 1) addresses a specific chain of challenges. First, to establish a stable global representation, we utilize the Normalized Image-Text Contrastive (N-ITC) loss. However, since global aggregation struggles to capture fine-grained attributes and is vulnerable to textual noise, we introduce the Gradient-Attention Similarity Score (GASS) mechanism to explicitly identify and filter token-level noise. Finally, to successfully unify these global and local mechanisms, we empirically found that existing forward-divergence matching methods become unstable under explicit token masking. Therefore, we propose

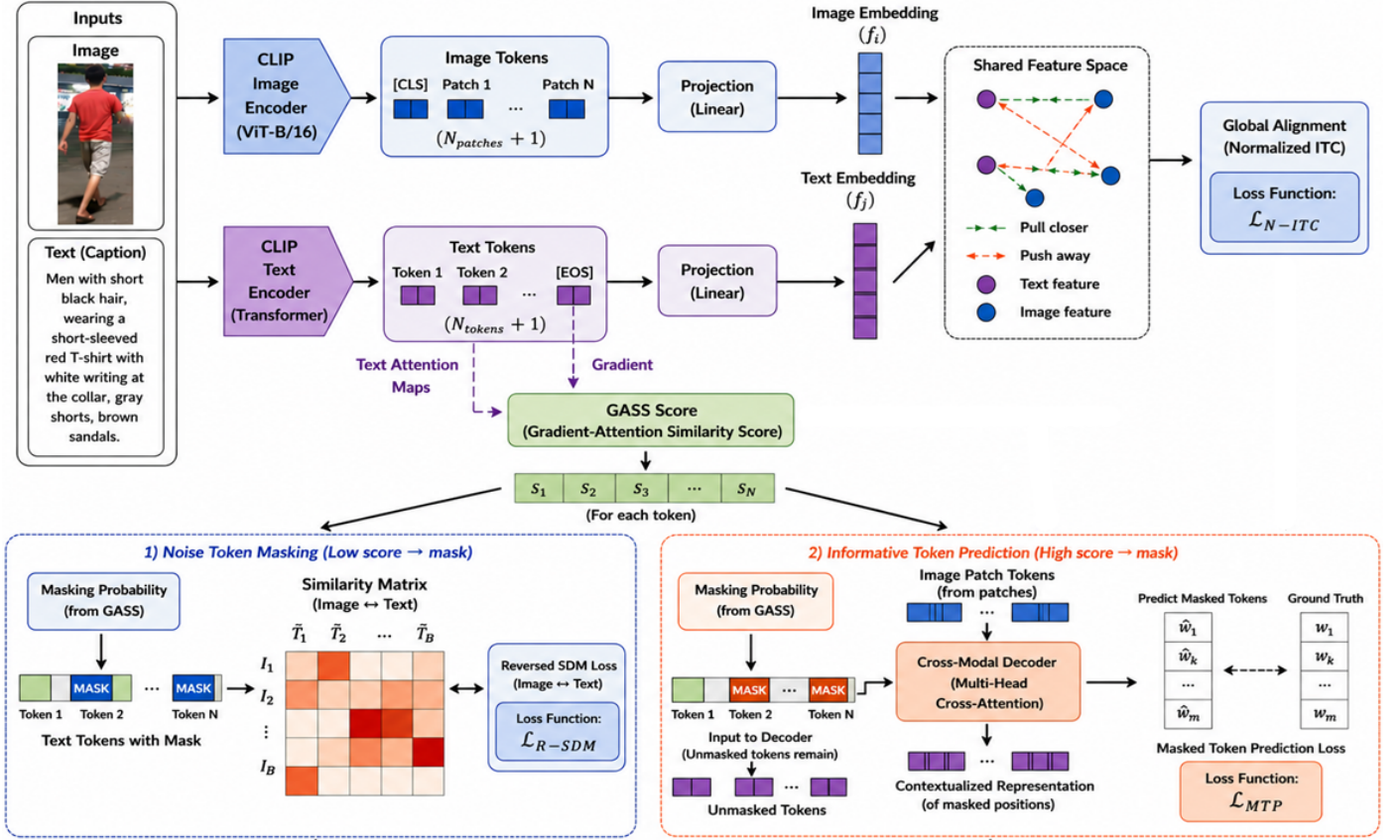


Fig. 1. Overview of our proposed NDM-CLIP framework

a novel reversed Similarity Distribution Matching (SDM) loss to optimize the fine-grained alignment robustly.

A. Feature Extraction and Global Alignment

The fundamental step of NDM-CLIP is to establish a reliable global correspondence between the raw caption and the image. Given an image-text pair (I, T) , we utilize the pre-trained CLIP dual-encoder architecture to extract representations. The image I is processed by a Vision Transformer (ViT) into a sequence of visual patch embeddings along with a global [cls] token. Concurrently, the text T is tokenized and fed into a Text Transformer to obtain sequential word embeddings and a global [eos] token. These global tokens are linearly projected into a shared multimodal space to form the visual and textual feature vectors, explicitly defined as f_{I_i} and f_{T_j} respectively.

While the standard CLIP model relies on an ordinary image-text contrastive loss on these vectors, this approach often suffers from embedding compression. To address this, we adopt the normalized variant proposed in [2]. Let their cosine similarity be $s(I_i, T_j) = f_{I_i}^T f_{T_j}$. The Normalized Image-Text Contrastive (N-ITC) loss explicitly levels the target distribution and is formulated as the symmetric sum of image-to-text and text-to-image alignments:

$$\mathcal{L}_{N-ITC} = \frac{1}{2} (\mathcal{L}_{I2T} + \mathcal{L}_{T2I}), \quad (1)$$

where the image-to-text component is computed across the mini-batch of size B as:

$$\mathcal{L}_{I2T} = -\frac{1}{B} \sum_{i=1}^B \sum_{j=1}^B q_{i,j}^* \log \left(\frac{\exp(s(I_i, T_j)/\tau)}{\sum_{k=1}^B \exp(s(I_i, T_k)/\tau)} \right). \quad (2)$$

Here, $q_{i,j}^* = q_{i,j} / \sum_{k=1}^B q_{i,k}$ serves as the row-normalized ground-truth label, mitigating the strict one-to-one constraint of traditional contrastive learning, and τ is a learnable temperature parameter. $\mathcal{L}_{T2I}$ is computed symmetrically. By directly balancing the target distribution, N-ITC provides a stable global embedding space, preparing the ground for local feature extraction.

B. Local Feature Filtering via Gradient-Attention

While N-ITC establishes a balanced global representation, its reliance on holistic sentence aggregation makes it highly susceptible to local textual noise, such as incorrect attributes or hallucinated words. To transition from global alignment to fine-grained local matching, the system must dynamically differentiate informative attributes from noisy tokens.

To achieve this efficiently, we integrate the Gradient-Attention Similarity Score (GASS) mechanism [3]. Instead of depending exclusively on standard self-attention, GASS extracts the gradient saliency of the [eos] token with respect

to the similarity score, and fuses it with textual self-attention maps across the final layers of the text encoder.

This mechanism yields a normalized score matrix $S \in \mathbb{R}^{B \times N}$ (where N is the sequence length) for each batch. A higher token score $s_k \in S$ indicates strong visual-semantic alignment, signifying an informative token, whereas a lower score identifies potential textual noise. In our framework, the matrix S serves as a deterministic token-level filter, guiding the dual-masking probabilities in our subsequent fine-grained optimization phase.

C. Fine-Grained Alignment with Reversed SDM

Guided by the filtering matrix S , GASS directs two complementary masking operations: noise suppression and informative token prediction.

1) *Noise Token Masking with Reversed SDM*: To mitigate textual noise, we compute a dropping probability for the k -th text token based on its GASS value s_k . We expand the probabilistic formulation as follows:

$$P_{\text{noise}}(s_k) = \frac{\alpha_n}{1 + \exp(-\lambda(1 - s_k - \gamma))}, \quad (3)$$

where α_n caps the maximum masking probability, while λ and γ control the steepness and midpoint of the distribution. During training, tokens are probabilistically substituted with [mask], yielding a noise-suppressed text representation $\tilde{T}_j$. Consequently, the updated matching probability $\tilde{p}_{i,j}$ relies on the similarities between the original image I_i and the masked text $\tilde{T}_j$.

The core novelty of our synergetic framework lies in how we align these noise-suppressed distributions. Standard approaches use the forward Kullback-Leibler divergence $D_{\text{KL}}(\mathbf{P} \parallel \mathbf{Q})$. However, under explicit token masking, the forward direction is fundamentally flawed as it scatters optimization gradients over many noisy false negatives. We resolve this by formulating a **reversed** SDM loss, minimizing $D_{\text{KL}}(\mathbf{Q} \parallel \mathbf{P})$. Expanding this divergence for the image-to-text direction, we obtain:

$$\mathcal{L}_{\text{R-SDM}}^{\text{I2T}} = \frac{1}{B} \sum_{i=1}^B \sum_{j=1}^B q_{i,j}^* (\log q_{i,j}^* - \log(\tilde{p}_{i,j} + \epsilon)), \quad (4)$$

with a small constant ϵ preventing numerical instability. This reversal directly anchors the optimization exclusively to the true positive probability mass of $q_{i,j}^*$, effectively insulating the training process from mask-induced noise. Combined with the symmetric text-to-image term, the final reversed matching objective is $\mathcal{L}_{\text{R-SDM}} = \mathcal{L}_{\text{R-SDM}}^{\text{I2T}} + \mathcal{L}_{\text{R-SDM}}^{\text{T2I}}$.

2) *Masked Informative Token Prediction*: To enforce the learning of local semantics, informative tokens are masked with probability $P_{\text{info}}(s_k) = \alpha_i / (1 + \exp(-\lambda(s_k - \gamma)))$. To reconstruct these tokens, we employ a cross-modal interaction decoder, which incorporates multi-head cross-attention layers to densely fuse the unmasked visual contexts into the textual queries. The hidden state h_k^m of a masked token at position $k \in \mathcal{M}$ (where $\mathcal{M}$ is the masked set) is then mapped to a predicted vocabulary distribution $\mathbf{z}_k$. The Masked Token

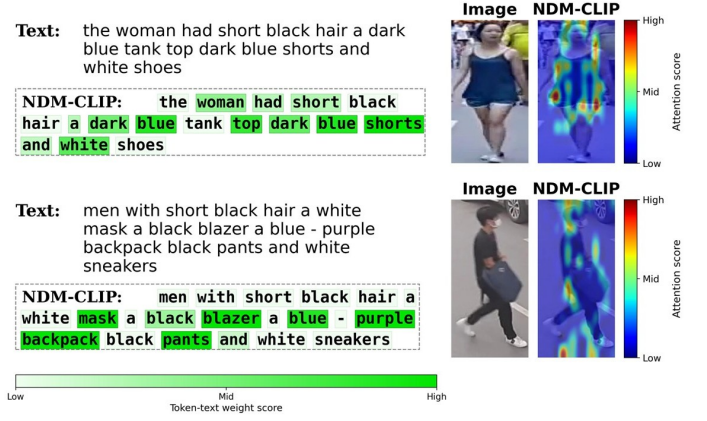


Fig. 2. Visualization of token-wise weight scores and attention maps. GASS token importance (left, darker green denotes higher scores) and corresponding NDM-CLIP attention heatmaps (right). The visualizations demonstrate precise localization of fine-grained visual attributes.

Prediction (MTP) loss is computed as the expected cross-entropy over the masked indices:

$$\mathcal{L}_{\text{MTP}} = \mathbb{E}_{k \in \mathcal{M}} \left[- \sum_{v=1}^{|\mathcal{V}|} y_{k,v} \log \left(\frac{\exp(z_{k,v})}{\sum_{c=1}^{|\mathcal{V}|} \exp(z_{k,c})} \right) \right], \quad (5)$$

where $|\mathcal{V}|$ is the vocabulary size and $y_{k,v}$ represents the one-hot encoded ground-truth target. The overall objective function integrates these components via a weighting factor β :

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{N-ITC}} + \mathcal{L}_{\text{R-SDM}} + \beta \mathcal{L}_{\text{MTP}}. \quad (6)$$

IV. EXPERIMENTS AND RESULTS

A. Datasets

We evaluate on two TBPS benchmarks with different linguistic characteristics: CUHK-PEDES [6] (English, high-resource) and 3000Vn-V2E [20] (English-translated). CUHK-PEDES contains 40,206 images of 13,000 identities; the standard split uses 34,054 training images (11,003 IDs) and 3,074 test images (1,000 IDs). 3000Vn-V2E is the English-translated version of the Vietnamese 3000VnPersonSearch dataset [21], with 4,302 training images and 2,000 test images.

B. Evaluation Metrics

Following common practice, we report Rank- k accuracy (R@1 , R@5 , R@10) and mean Average Precision (mAP).

C. Implementation Details

NDM-CLIP is implemented in PyTorch using the CLIP ViT-B/16 backbone with input images resized to 224×224 . Training proceeds for 15 epochs with a batch size of 16 on an NVIDIA RTX 3060 GPU, using AdamW with a learning rate of 1×10^{-5} . The empirically tuned hyperparameters are: $\beta = 0.5$; for masking, $\alpha_n = 0.2$, $\alpha_i = 0.3$, $\lambda = 10$, $\gamma = 0.5$, and GASS is computed from the last $L = 8$ layers.



Fig. 3. Qualitative top-5 retrieval results on 3000Vn-V2E. Correct matches are shown in green, incorrect ones in red.

D. Ablation Study

1) Reversed vs. Forward SDM: Table I demonstrates that switching to the reversed $D_{KL}(Q||P)$ brings improvement over the forward version. Under heavy token masking, forward KL scatters gradients across many false negatives, destabilizing training. In contrast, reversed SDM anchors gradients to true labels, effectively insulating optimization from masking-induced noise.

TABLE I
ABLATION STUDY ON A 10% CUHK-PEDES FOLD. ‘F’ IS FORWARD $D_{KL}(P||Q)$, ‘R’ IS REVERSED $D_{KL}(Q||P)$.

N-ITC	SDM (F)	SDM (R)	MTP	R@1	R@5	R@10	mAP
	✓		✓	45.11	69.14	78.36	41.69
		✓	✓	48.54	72.06	79.81	43.13
✓	✓		✓	45.45	69.66	79.04	41.86
✓		✓		50.24	73.41	81.68	44.75
✓		✓	✓	51.72	75.08	82.36	46.06

2) Component Synergy: Removing N-ITC drops R@1 from 51.72% to 48.54%, confirming its role in balancing the global manifold before local matching. Disabling MTP reduces R@1 to 50.24%, underscoring the importance of fine-grained token reconstruction.

E. Comparison with State-of-the-Art Methods

1) Data-Limited Regime (10% CUHK-PEDES): We trained NDM-CLIP on 10% random subsets over ten folds (Table II); the mean R@1 is 51.81% with a standard deviation of

only 0.66%. Compared with other methods under the same data constraint (Table III), NDM-CLIP outperforms TBPS-CLIP by +16.32% and TBPS-mSigLIP by +5.08%.

TABLE II
RESULTS ON 10% CUHK-PEDES (10 RANDOM FOLDS).

Fold	R@1	R@5	R@10	mAP
1	51.75	74.01	81.73	45.74
2	52.41	74.65	82.20	46.21
3	51.05	73.12	80.85	44.75
4	52.64	74.82	82.45	46.55
5	51.52	73.80	81.50	45.40
6	52.12	74.35	81.95	46.90
7	50.95	72.90	80.70	44.50
8	51.35	73.55	81.25	45.15
9	52.83	75.15	82.72	46.85
10	51.46	74.30	82.48	46.00
Mean	51.81	74.07	81.78	45.81
Std. Dev.	0.66	0.73	0.70	0.85

TABLE III
COMPARISON ON 10% CUHK-PEDES DATASET.

Method	R@1	R@5	R@10	mAP
ViTAA [4]	31.27	54.96	65.38	—
TBPS-CLIP [2]	35.49	59.93	69.33	34.41
TBPS-mSigLIP [19]	46.73	68.65	77.55	41.75
NDM-CLIP (Ours)	51.81	74.07	81.78	45.81

2) Robustness on Translated Queries (3000Vn-V2E): On the machine-translated English queries (Table IV),

NDM-CLIP achieves 58.81% R@1, surpassing TBPS-mSigLIP by +19.48%. This confirms that GASS-guided dual masking effectively filters the lexical inconsistencies caused by machine translation.

TABLE IV
RESULTS ON 3000VN-V2E.

Method	R@1	R@5	R@10	mAP
TBPS-CLIP [2]	35.40	62.45	73.93	41.76
M-IRRA [22]	37.07	63.30	74.25	42.44
TBPS-mSigLIP [19]	39.33	68.23	77.83	45.17
NDM-CLIP (Ours)	58.81	80.33	87.72	60.84

3) Full CUHK-PEDES Benchmark: Table V reports results on the complete dataset. To ensure a fair comparison under constrained computational resources, we re-trained representative baselines (denoted with ‘*’) using the same hardware configuration described in Section IV-C. Under smaller batch sizes, the Rank@1 performance of GA-DMS* and TBPS-CLIP* drops to 64.47% and 66.49%, respectively, highlighting their dependence on large-scale training conditions. In contrast, NDM-CLIP* remains robust, achieving 69.76% Rank@1 under the same setting. This stable performance suggests that the integration of GASS-guided masking, the coarse-to-fine learning strategy, and the reversed SDM objective effectively reduces the typical reliance on large batch sizes.

TABLE V
RESULTS ON FULL CUHK-PEDES.

Method	R@1	R@5	R@10	mAP
ViTAA [4]	55.97	75.84	83.52	—
SSAN [23]	61.37	80.15	86.73	—
CLIP (ViT-B/16) [1]	65.37	85.83	91.59	59.42
UniPT [10]	68.50	84.67	90.38	—
TBPS-mSigLIP [19]	70.76	87.44	92.06	63.40
Simplified TBPS-CLIP [2]	72.66	88.14	92.72	64.97
TBPS-CLIP [2]	73.54	88.19	92.35	65.38
IRRA [9]	73.38	89.93	93.71	66.13
GA-DMS [3]	77.60	91.40	94.78	69.82
TBPS-CLIP*	66.49	84.47	90.41	59.01
GA-DMS*	64.47	84.19	90.20	57.95
NDM-CLIP* (Ours)	69.76	86.47	91.37	61.09

V. CONCLUSION

We proposed NDM-CLIP, a synergistic framework that advances Text-Based Person Search by combining N-ITC, dynamic GASS-guided masking, and Masked-Text Prediction. A crucial contribution of our work is resolving the gradient instability inherent in standard forward KL divergence under explicit noise masking. By introducing a reversed $D_{KL}(Q||P)$ SDM objective, we ensure the optimization process remains strictly anchored to ground-truth labels. Comprehensive experiments validate that our approach significantly outperforms existing methods, demonstrating exceptional data efficiency in data-scarce settings and robust performance in noisy, real-world textual environments.

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